

PROVEIT RESEARCH CONTINUATION

Signed Condensation in Quadratic-Feedback Transseries

Sharp reversion, universal cancellation,
and inverse Borel growth

Research report prepared for Vladimir Reshetnikov

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Abstract

For the formal solution of $\widehat{U}(q) = \sum_{j \geq 1} w_j q^j \exp(\lambda_j \widehat{U}(q))$, suppose that $w_1 = 1$, all primitive weights and slopes are nonnegative, and, outside a finite prefix, $w_j = 1$ and $\lambda_j = aj^2 + bj + d$, with $a > 0$. We determine a full multiplicative equivalent for the coefficients of its compositional inverse. With $\beta = \lambda_1 + w_2$, $r_n(1 + r_n)e^{2r_n} = an$, and $S_n = \exp(nr_n(1 + 2r_n)/(1 + r_n))/\sqrt{1 + 4r_n + 2r_n^2}$, the result is

$$[u^n]\widehat{Q}(u) \sim -S_n \exp\left(\frac{b - \beta}{a}r_n\right).$$

For $w_j = 1$ and $\lambda_j = aj^2$, this proves the precise quadratic inverse conjecture in the repository's finite-core article. The essential step is a signed-tail localization theorem: after a sufficiently large but fixed analytic core is removed, configurations containing at least two remaining actions contribute less than $S_n n^{-A}$, for any prescribed $A > 0$. It is proved at the level of absolute configuration weights, before cancellations are used. We then derive eventual negativity, a finite-prefix sensitivity law, the exact positive-axis growth equivalent of the inverse Borel transform, and the least inverse-term equivalent. The proofs are self-contained apart from classical inversion and elementary asymptotic tools. An exact verification program, recorded integer coefficients, a source audit, and a research agenda accompany the manuscript.

Status and scope. This is a research manuscript with proofs for independent review, not a Lean-verified development. The novelty claim is scoped to the explicitly audited repository conjecture and the extensions proved here; no claim of resolving general transseries summability or a classical field-wide conjecture is made.

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1 The specific unresolved problem

1.1 Repository provenance

The repository *ProveIt* separates its formal transseries library from its mathematical research reports. At the inspected snapshot, the project README expressly records that the recent research arrivals have not been reviewed claim by claim and do not become Lean theorems merely by being filed alongside the library [1, 2]. We retain that distinction. The fixed snapshot used throughout this report is

aab173a7ccdd48add72191d68ce6b0c12ffd72fe.

The principal source is

Analysis/Transseries/docs/series-and-transseries/
Finite_Core_Universality_Exponential_Feedback/article.tex.

Its blob identifier is

c2b6faf623003111e9d9e6bbb162a67429f3ce42.

The source’s Section “Quadratic and subquadratic inverse asymptotics,” particularly labels `prop:inverse-response`, `conj:quadratic-inverse`, and `eq:inverse-conjecture`, states an exact first-order inverse response but leaves the cancellation estimate needed to pass from that response to the full inverse unproved [3]. The relevant inspected source range is lines 1623–1678; the precise conjecture is at lines 1661–1668, with its explanation immediately after it.

For the pure quadratic model, that conjecture is

$$v_n \sim -S_n e^{-(1+1/a)r_n}, \quad \widehat{Q}(u) = \sum_{n \geq 1} v_n u^n. \quad (1.1)$$

The same article proves a forward equivalent $u_n \sim S_n e^{(1+1/a)r_n^2}$, but proves $v_n \sim -u_n$ only in its superquadratic regime. Consequently, Gevrey-class preservation, preservation of weighted exponential type, or an estimate of $\log |v_n|$ to accuracy $o(n)$ does not settle (1.1). In the quadratic case the missing factor in the forward-to-inverse comparison is already exponential in $(\log n)^2$.

1.2 What is added here

The principal contribution is a proof of (1.1), including its sign and multiplicative constant. We prove it in the larger class with an arbitrary nonnegative finite prefix and an eventually quadratic polynomial tail. The argument has two distinct components. A positive majorant proves that, after a sufficiently large fixed cutoff, exactly one remaining action accounts for the inverse coefficient up to an arbitrarily small polynomial error relative to S_n . A separate analytic-core calculation then sums the cancellations within that one-action contribution.

This separation matters. Bounding all inverse-composition terms absolutely at the start and comparing them directly with the desired answer loses the factor we are trying to determine. Conversely, computing a compelling one-action saddle does not exclude a larger contribution from the rest of the signed expansion. Both issues must be resolved.

Two further consequences contain more information than coefficient type: a multiplicative equivalent for the inverse Borel transform on the positive real axis, and a multiplicative equivalent for its smallest formal term. The latter is *not* asserted to be an analytic remainder estimate.

1.3 Relation to the literature

Lagrange inversion and its multivariate forms are classical; Gessel's survey is a convenient reference [4]. General inversion with infinitely many colours is developed by Jansen, Kuna, and Tsagkarogiannis [5]. Our finite coefficient formula is a specialization of that classical machinery, not a new inversion formula.

Borinsky develops an algebraic calculus for coefficients possessing fixed factorial asymptotic expansions and proves composition and inversion rules in that setting [6]. The present sequences have zero type relative to every fixed exponential multiple of $n!$, while still having zero ordinary radius of convergence. Thus a fixed-factorial expansion with all asymptotic coefficients zero cannot distinguish the cancellation laws proved below. This is a reason to keep the slowly varying saddle coordinate as part of the asymptotic data. The general formal transseries viewpoint is explained by Edgar [7]; our support is only $\{e^{-nx} : n \geq 1\}$. The difficulty lies in large-order coefficients, not in a new support order.

2 Model and principal theorems

Fix real numbers $a > 0$, $b, d \in \mathbb{R}$, and an integer $J_0 \geq 2$. Assume

$$w_1 = 1, \quad w_j \geq 0, \quad \lambda_j \geq 0, \quad w_j = 1, \quad \lambda_j = aj^2 + bj + d \quad (j > J_0). \quad (2.1)$$

All exceptional weights and slopes are finite. In particular, a weight in the finite prefix may vanish. Define

$$\widehat{U}(q) = \sum_{j \geq 1} w_j q^j e^{\lambda_j \widehat{U}(q)}, \quad \widehat{Q} = \widehat{U}^{(-1)}, \quad \beta = \lambda_1 + w_2. \quad (2.2)$$

These equations are formal in $\mathbb{R}[[q]]$. The coefficient of degree n in the defining equation only uses $j \leq n$ and earlier coefficients of $\widehat{U}$. Hence $\widehat{U}$ exists uniquely, $u_1 = 1$, and its formal inverse exists uniquely. Direct expansion gives

$$u_2 = \beta, \quad v_2 = -\beta. \quad (2.3)$$

For real $n > 0$, let $r = r(n) > 0$ solve

$$r(1+r)e^{2r} = an. \quad (2.4)$$

The left side is strictly increasing from zero to infinity. Put

$$j_n = \frac{n}{1+r_n}, \quad k_n = \frac{nr_n}{1+r_n}, \quad D_n = 1 + 4r_n + 2r_n^2, \quad S_n = \frac{e^{k_n(1+2r_n)}}{\sqrt{D_n}}. \quad (2.5)$$

Here j_n is a real saddle coordinate, not an integer index. We have $r_n \sim \frac{1}{2} \log n$, $j_n \sim 2n/\log n$, and $k_n \sim n$. The exact coordinate r_n , rather than a truncated logarithmic expansion, is essential in a multiplicative equivalent.

Theorem 2.1 (Sharp quadratic reversion). *Under (2.1),*

$$\boxed{v_n \sim -S_n \exp\left(\frac{b-\beta}{a} r_n\right)}. \quad (2.6)$$

In particular, $v_n < 0$ for every sufficiently large integer n . The constant term d of the tail affects neither the displayed exponential factor nor its leading multiplicative constant.

Corollary 2.2 (The repository conjecture). *For $w_j = 1$, $\lambda_j = aj^2$ at every $j \geq 1$,*

$$v_n \sim -\frac{\exp\left(\frac{nr_n(1+2r_n)}{1+r_n} - (1+1/a)r_n\right)}{\sqrt{1+4r_n+2r_n^2}}.$$

Thus (1.1) holds for every fixed $a > 0$.

Corollary 2.3 (Finite-prefix sensitivity). *Suppose two models have the same eventual a, b, d and respective values $\beta, \tilde{\beta}$. Then*

$$\frac{v_n}{\tilde{v}_n} \sim \exp\left(-\frac{\beta - \tilde{\beta}}{a}r_n\right). \quad (2.7)$$

In particular, preserving u_2 makes every other finite-prefix alteration invisible in the leading inverse equivalent.

For the remaining statements, write $\gamma = (b - \beta)/a$ and use the ordinary factorial-normalized Borel transform

$$\mathcal{B}\hat{Q}(z) = \sum_{n \geq 1} \frac{v_n}{n!} z^n. \quad (2.8)$$

Theorem 2.4 (Inverse Borel growth). *The series (2.8) is entire. For positive real $z \rightarrow \infty$, let $R = R(z) > 0$ be defined by $R(R+1) = az$. Then*

$$\mathcal{B}\hat{Q}(z) \sim -\frac{e^{\gamma R}}{\sqrt{2R+1}} \exp\left(\frac{R}{a}e^{2R}\right). \quad (2.9)$$

Consequently its ordinary Borel–Laplace integral in the positive direction does not exist, for any positive Laplace parameter.

Theorem 2.5 (Least inverse term). *Choose N so large that $v_n < 0$ for $n \geq N$, and set $R = \frac{1}{2} \log(1/t)$ for $t \downarrow 0$. Then*

$$\min_{n \geq N} |v_n| t^n \sim \frac{\exp\left(-\frac{R^2}{at} + \gamma R\right)}{\sqrt{1+4R+2R^2}}. \quad (2.10)$$

Every minimizing index satisfies

$$n_{\min} = \frac{R(R+1)}{at} + o\left(\sqrt{\frac{R(R+1)}{at}}\right). \quad (2.11)$$

No claim about a uniquely minimizing integer, an $O(1)$ index error, or an analytic truncation remainder is included.

3 An exact signed configuration formula

The inverse can be defined directly, without first computing $\hat{U}$:

$$u = \sum_{j \geq 1} w_j \hat{Q}(u)^j e^{\lambda_j u}. \quad (3.1)$$

Put $\hat{Q}(u) = ue^{-\lambda_1 u} Z(u)$. Then

$$1 = Z + \sum_{j \geq 2} x_j Z^j, \quad x_j = w_j u^{j-1} e^{(\lambda_j - j\lambda_1)u}. \quad (3.2)$$

Lemma 3.1 (Finite coefficient formula). *For a finitely supported multiplicity vector $\mathbf{m} = (m_j)_{j \geq 2}$, write*

$$K = \sum_{j \geq 2} m_j, \quad L = \sum_{j \geq 2} (j-1)m_j, \quad h = n-1-L, \quad E(\mathbf{m}) = \sum_{j \geq 2} (\lambda_j - j\lambda_1)m_j - \lambda_1. \quad (3.3)$$

Then

$$v_n = \frac{(-\lambda_1)^{n-1}}{(n-1)!} + \sum_{\substack{\mathbf{m} \neq 0 \\ L \leq n-1}} (-1)^K \frac{(L+K)!}{(L+1)! \prod_j m_j!} \left(\prod_j w_j^{m_j} \right) \frac{E(\mathbf{m})^h}{h!}. \quad (3.4)$$

The sum is finite. We use $0^0 = 1$, as required by the exponential series.

Proof. Set $Y = 1 - Z$ in (3.2), and temporarily replace every x_j by sx_j . The equation is $Y = s \sum_j x_j (1 - Y)^j$. Ordinary Lagrange inversion in s gives

$$[s^K]Y = \frac{1}{K} [t^{K-1}] \left(\sum_j x_j (1-t)^j \right)^K.$$

For a specified multiplicity vector with $K \geq 1$, the multinomial factor is $K! / \prod m_j!$, and the exponent of $1-t$ is $\sum j m_j = L + K$. Thus its coefficient in $Z = 1 - Y$ is

$$-\frac{(K-1)!}{\prod m_j!} (-1)^{K-1} \binom{L+K}{K-1} = (-1)^K \frac{(L+K)!}{(L+1)! \prod m_j!}.$$

Substitute the x_j , multiply by $ue^{-\lambda_1 u}$, and take the coefficient of u^n . This yields (3.4). The cost $j-1 \geq 1$ forces $j \leq n$, $K \leq L \leq n-1$, proving finiteness. $\square$

For the pure model, the exponent simplifies to

$$E(\mathbf{m}) = a \left(\sum_{j \geq 2} j(j-1)m_j - 1 \right) > 0 \quad (\mathbf{m} \neq 0). \quad (3.5)$$

Therefore the signs in its nonempty configuration sum are exactly $(-1)^K$. The initial term in (3.4) is kept separate. For a general finite prefix, E can have either sign at small degree; the localization proof below uses its absolute value until a large action makes it positive.

Definition 3.2 (Absolute configuration mass). Let $A_{n,L}$ be the sum of the absolute values of the nonempty terms in (3.4) with the specified excess L . Let A_n be their sum over L . For an integer $M \geq J_0$, let $A_{n,M}^{(2)}$ be the same absolute mass restricted to configurations having at least two primitive actions $j > M$, counted with multiplicity.

These masses are not the absolute value of an already cancelled coefficient. They are larger, positive quantities. The next two sections prove estimates for them strong enough to justify later signed manipulations.

4 The bare saddle and its weighted sum

Define, for integers $1 \leq j \leq n$,

$$T_{n,j} = \frac{(aj^2)^{n-j}}{(n-j)!}, \quad F_n(x) = (n-x) \left(1 + \log \frac{ax^2}{n-x} \right) \quad (0 < x < n). \quad (4.1)$$

Differentiation gives

$$F'_n(x) = -\log \frac{ax^2}{n-x} + \frac{2(n-x)}{x}, \quad (4.2)$$

$$F''_n(x) = -\frac{4}{x} - \frac{1}{n-x} - \frac{2(n-x)}{x^2} < 0. \quad (4.3)$$

Thus the real saddle is unique. Equations (2.4)–(2.5) give

$$F'_n(j_n) = 0, \quad F_n(j_n) = k_n(1 + 2r_n), \quad -F''_n(j_n) = \frac{D_n}{k_n}, \quad \sigma_n^2 = \frac{k_n}{D_n} \asymp \frac{n}{(\log n)^2}. \quad (4.4)$$

Lemma 4.1 (Bare and slowly weighted saddle sums). *For every fixed real c ,*

$$\sum_{j=1}^n T_{n,j} \exp\left(c \frac{n-j}{j}\right) \sim S_n e^{cr_n}. \quad (4.5)$$

The sum may be restricted to $n/\log n \leq j \leq 3n/\log n$. The discarded part is exponentially small relative to the full sum. Also, in a fixed sufficiently small relative neighborhood of j_n ,

$$F_n(j_n) - F_n(j) \geq c_0 \frac{(\log n)^2}{n} (j - j_n)^2 \quad (4.6)$$

for a fixed $c_0 > 0$ and all sufficiently large n .

Proof. Stirling's formula gives, uniformly on any compact positive multiple of $n/\log n$,

$$T_{n,j} = \frac{e^{F_n(j)}}{\sqrt{2\pi(n-j)}} (1 + O(n^{-1})).$$

On such an interval, $-F''_n(j) \asymp (\log n)^2/n$, proving (4.6). The third derivative is $O((\log n)^3/n^2)$, so on $j = j_n + \sigma_n y$, with bounded y ,

$$F_n(j) = F_n(j_n) - y^2/2 + O(n^{-1/2}|y|^3).$$

Moreover, $(n-j)/j = r_n + O((\log n)/\sqrt{n})$ on every bounded standardized interval. The Gaussian Riemann sum, whose mesh is $1/\sigma_n \rightarrow 0$, therefore contributes

$$\frac{e^{F_n(j_n)+cr_n}}{\sqrt{2\pi k_n}} \sqrt{2\pi} \sigma_n = S_n e^{cr_n}.$$

The quadratic bound dominates larger standardized intervals within the fixed relative neighborhood. On any other compact multiple of $n/\log n$, the scaled phase has a strict maximum at 2: with $y = j \log n/n$,

$$\frac{F_n(j)}{n} = \log n - 2 \log \log n + \log a + 1 + 2 \log y - y + o(1).$$

This gives a fixed exponential loss outside a neighborhood of $y = 2$. For j/n bounded below, the leading $n \log n$ coefficient is smaller; for smaller j , the same expression and the limit $2 \log y - y \rightarrow -\infty$ exclude the remaining range. These bounds also absorb the fixed weight $e^{c(n-j)/j}$ away from the saddle: for very small j it contributes at most $e^{|c|n}$, whereas the main logarithm is $n \log n - O(n \log \log n)$. Strict concavity handles the intermediate ranges. This proves the claims. $\square$

Remark 4.2. The determinant factor $D_n^{-1/2}$ is the product of the factorial prefactor and the width of the discrete action saddle. Omitting either factor changes the multiplicative constant. Replacing r_n in the large exponential by $\frac{1}{2} \log n$ does not preserve an equivalent.

5 Absolute localization before cancellation

Theorem 5.1 (Arbitrarily accurate fixed-core localization). *Under (2.1), there is a constant $C > 0$ such that*

$$A_n \leq S_n n^C \quad \text{for all sufficiently large } n. \quad (5.1)$$

For every $A > 0$, there is a fixed integer $M \geq J_0$ such that

$$A_{n,M}^{(2)} \leq S_n n^{-A} \quad \text{for all sufficiently large } n. \quad (5.2)$$

The integer M may depend on the model and on A , but not on n .

The proof is given in several steps to make the absolute error scale explicit. Constants below depend on the fixed model unless stated otherwise. They never depend on the configuration or on n . Constants carrying a subscript M are permitted to depend on the fixed cutoff.

5.1 A row bound and macroscopic localization

Write the excesses of the nonunit actions as $\ell_i = j_i - 1 \geq 1$, so $L = \sum_i \ell_i$. The eventual polynomial hypothesis implies

$$|E(\mathbf{m})| \leq a \sum_i \ell_i^2 + C_1 L + C_1 \leq a L^2 + C_1 L + C_1. \quad (5.3)$$

Choose $W \geq 1$ bounding all weights. For a fixed L and number K of nonunit actions,

$$\sum_{\substack{\mathbf{m}: \sum m_j = K \\ \sum (j-1)m_j = L}} \frac{1}{\prod m_j!} = \frac{1}{K!} \binom{L-1}{K-1}.$$

Consequently the total combinatorial factor is at most

$$\sum_{K=1}^L \frac{(L+K)!}{(L+1)!K!} W^K \binom{L-1}{K-1} \leq (8W)^L. \quad (5.4)$$

For example, bound $\binom{L+K}{K} \leq 4^L$, $\binom{L-1}{K-1} \leq 2^L$, and use $L/(L+1) \leq 1$. It follows that

$$A_{n,L} \leq (8W)^L \frac{(aL^2 + C_1 L + C_1)^{n-1-L}}{(n-1-L)!}. \quad (5.5)$$

Lemma 5.2 (Macroscopic absolute localization). *Fix sufficiently small $\delta, \varepsilon > 0$. Except for absolute mass at most $S_n e^{-cn}$, with some $c > 0$, every configuration satisfies*

$$|L + 1 - j_n| \leq \delta j_n, \quad \ell_{\max} \geq (1 - \varepsilon)L. \quad (5.6)$$

The largest excess is unique when $\varepsilon < 1/2$.

Proof. We detail the localization implicit in (5.5). A row with $L \geq \eta n$, for fixed $\eta > 0$, has logarithm at most $(1 - \eta)n \log n + O(n \log \log n + n)$, so cannot approach the bare maximum. For rows within $O(n)$ of that maximum, necessarily $L/n \rightarrow 0$ and $\log L = \log n - O(\log \log n)$: otherwise the base $aL^2 + O(L + 1)$, divided by a factorial with index asymptotic to n , loses a positive multiple of $n \log \log n$ or $n \log n$. In this restricted range, logarithmic expansion of the base and Stirling's formula give the same leading variational expression as in Lemma 4.1, with $y = L \log n/n$:

$$\frac{1}{n} \log A_{n,L} \leq \log n - 2 \log \log n + \log a + 1 + 2 \log y - y + o(1). \quad (5.7)$$

Here the entropy $O(L)$, the linear correction in the base, and the factorial error are all $o(n)$. One can first force $y = O(\log \log n)$ by comparison with the maximum, so the expansion is valid along every potentially maximizing sequence; then $2 \log y - y \rightarrow -\infty$ at both ends forces y into a compact subset of $(0, \infty)$. Its unique strict maximum is at $y = 2$. This proves a fixed exponential loss outside the first condition of (5.6); $j_n \log n/n \rightarrow 2$. Summing the at most n rows only contributes $O(\log n)$ to their logarithm.

If $\ell_{\max} \leq (1 - \varepsilon)L$, convexity, or successively transferring excess between two nonextreme parts, gives

$$\sum_i \ell_i^2 \leq ((1 - \varepsilon)^2 + \varepsilon^2)L^2.$$

Integer rounding changes this by $O(L)$, which is harmless. The factor in parentheses is strictly less than one. Replacing a by this factor times a in (5.7) lowers the maximum by a fixed positive multiple of n . This proves the second assertion and the combined exponential estimate. Uniqueness follows because two excesses larger than $(1 - \varepsilon)L$ cannot sum to at most L . $\square$

5.2 The cost of a background cloud

In the region (5.6), write

$$\ell_{\max} = L - B, \quad B = \ell_1 + \cdots + \ell_s \leq \varepsilon L,$$

where $s = K - 1$ counts the remaining actions. The largest action is in the unperturbed tail for all sufficiently large n . Fuse the entire background into it, obtaining a single action $L + 1$. Its exponent is

$$E_*(L) = \lambda_{L+1} - (L + 2)\lambda_1 = a(L + 1)^2 + (b - \lambda_1)(L + 1) + d - \lambda_1. \quad (5.8)$$

This quantity is positive and asymptotic to aL^2 . The difference in exponents obeys

$$E_*(L) - E(\mathbf{m}) \geq 2aB(L - B) - C_2B. \quad (5.9)$$

To check the finite-prefix issue explicitly, write $\lambda_j = aj^2 + bj + d + \eta_j$, where $\eta_j = 0$ outside the prefix. The quadratic part of the difference is $a(L^2 - \sum_i \ell_i^2)$, and every other term in the difference is bounded in absolute value by a constant times $s \leq B$. Since $\sum_{i=1}^s \ell_i^2 \leq B^2$, (5.9) follows. In this region $E(\mathbf{m}) > 0$ as well.

For $h = n - 1 - L$, (5.9) implies

$$\left(\frac{E(\mathbf{m})}{E_*(L)} \right)^h \leq \exp \left[- \left(2 - \frac{2B}{L} - \frac{C_3}{L} \right) \frac{hB}{L} \right]. \quad (5.10)$$

The combinatorial coefficient relative to the single fused action is

$$\frac{(L + s + 1)!}{(L + 1)! \prod_{j \neq j_{\max}} m_j!}.$$

After replacing multiplicities by ordered background excesses and dividing by $s!$, it is at most $(2WL)^s/s!$, including weights, because $s \leq B \leq \varepsilon L$. Thus, whenever $2 - 2B/L - C_3/L \geq \kappa > 0$, the total relative background mass is at most

$$\sum_{s \geq 0} \frac{(2WL)^s}{s!} \left(\sum_{\ell \geq 1} e^{-\kappa h \ell / L} \right)^s = \exp \left(\frac{2WL}{e^{\kappa h / L} - 1} \right). \quad (5.11)$$

Restrictions such as a prescribed total B or an upper bound on each background excess were dropped only to increase this positive bound.

5.3 A finer excess window

Take, for definiteness, fixed sufficiently small $\delta, \varepsilon \leq 1/100$ in Lemma 5.2. For all sufficiently large n we can use $\kappa = 2 - 4\varepsilon$ in (5.11). Uniformly in the macroscopic excess window,

$$\frac{2WL}{e^{\kappa h/L} - 1} \leq n^{1/3}. \quad (5.12)$$

Indeed, $L \asymp n/\log n$ and $h/L \geq r_n/(1 + \delta) + O(1)$; the power of n on the left is strictly smaller than $1/3$ for these choices. Moreover

$$\frac{E_*(L)^h}{h!} = T_{n,L+1} \exp(O(\log n)) \quad (5.13)$$

uniformly in that window. By (4.6), configurations outside

$$|L + 1 - j_n| \leq \frac{j_n}{(\log n)^2} \quad (5.14)$$

but within (5.6) therefore have total mass at most

$$S_n \exp\left(-c_1 \frac{n}{(\log n)^4} + n^{1/3} + O(\log n)\right) \leq S_n \exp\left(-c_2 \frac{n}{(\log n)^4}\right). \quad (5.15)$$

This refinement is why an $e^{o(n)}$ majorant alone is not enough: we need to reach a window in which the remaining cloud costs only a power of n .

5.4 Excluding a mesoscopic background

Inside (5.14), consider $B \geq L/\log n$, still with $B \leq \varepsilon L$. Split the exponent in the coarse cloud bound in half. The resulting total background mass, relative to $E_*(L)^h/h!$, is at most

$$\exp\left(-\frac{\kappa h}{2 \log n}\right) \exp\left(\frac{2WL}{e^{\kappa h/(2L)} - 1}\right) \leq \exp\left(-c_3 \frac{n}{\log n}\right). \quad (5.16)$$

For the last inequality, the positive logarithm is $O(n^{1-\kappa/4+o(1)}) = o(n/\log n)$, whereas $h \sim n$. This is a direct exponential-moment bound, not a probabilistic independence assumption.

5.5 The polynomial cloud and marked-tail estimate

It remains to consider (5.14) and $B < L/\log n$. Now $h/L = r_n + O(1/\log n)$, and (5.10) permits

$$z_n(L) = \exp\left[-\left(2 - \frac{2}{\log n} - \frac{C_3}{L}\right) h/L\right] \leq C_4 e^{-2r_n}.$$

The cloud exponent satisfies

$$\mu_n(L) = \frac{2WLz_n(L)}{1 - z_n(L)} \leq C_5 r_n, \quad (5.17)$$

since $e^{-2r_n} = r_n(1 + r_n)/(an)$ and $L \sim n/(1 + r_n)$. Together with (5.13) and Lemma 4.1, this proves (5.1), including the previously discarded exceptions.

If there is a second primitive action $j > M$, one background excess is at least M . Marking one such occurrence in the exponential majorant bounds the relative mass by

$$e^{\mu_n(L)} \frac{2WLz_n(L)^M}{1 - z_n(L)}. \quad (5.18)$$

The marked count is at least one on the event being bounded, so multiple marks only overcount. Combining (5.18) with the reference row estimate yields, for every fixed $M \geq J_0$,

$$A_{n,M}^{(2)} \leq C_M S_n r_n \exp(-(2M - 2 - C_6)r_n) + S_n \exp\left(-c_4 \frac{n}{(\log n)^4}\right). \quad (5.19)$$

Here C_6 is independent of M ; factors such as C_4^M are absorbed in C_M . The macroscopic exceptional set includes configurations with no unique large action, so none have been omitted. Because $r_n \sim \frac{1}{2} \log n$, taking M sufficiently large proves (5.2) for any prescribed A . This completes the proof of Theorem 5.1. $\square$

Remark 5.3 (What this theorem does not optimize). The proof gives existence of a sufficient fixed cutoff and arbitrary polynomial accuracy. It does not identify the smallest sufficient cutoff. In particular, it should not be confused with the sharp positive-coefficient cutoff in the repository's forward theorem. Improving C_6 or proving that a much smaller inverse core always suffices remains a separate problem.

6 The analytic core sums the cancellations

Fix $M \geq J_0$. Define

$$F_M(q, u) = \sum_{j=1}^M w_j q^j e^{\lambda_j u}, \quad F_M(Q_M(u), u) = u, \quad A_M(u) = \frac{1}{\partial_q F_M(Q_M(u), u)}. \quad (6.1)$$

The analytic implicit-function theorem applies because $\partial_q F_M(0, 0) = 1$. Thus Q_M, A_M are analytic in a fixed complex disc around zero. Their initial terms are

$$Q_M(u) = u - \beta u^2 + O(u^3), \quad A_M(u) = 1 + O(u). \quad (6.2)$$

Lemma 6.1 (Exact one-tail decomposition). *Define the formal series*

$$V_M^{(1)}(u) = -A_M(u) \sum_{j>M} Q_M(u)^j e^{\lambda_j u}. \quad (6.3)$$

Then

$$\hat{Q}(u) = Q_M(u) + V_M^{(1)}(u) + R_M(u), \quad |[u^n]R_M(u)| \leq A_{n,M}^{(2)}. \quad (6.4)$$

Moreover, restricting the one-tail sum to $n/\log n \leq j \leq 3n/\log n$ changes its degree- n coefficient by $O(S_n e^{-cn})$, for a suitable fixed $c > 0$.

Proof. Give every primitive action $j > M$ a marker s , and let Q_s be the formal inverse defined by

$$u = F_M(Q_s, u) + s \sum_{j>M} Q_s^j e^{\lambda_j u}.$$

Every fixed u -coefficient is a polynomial in s . At $s = 0$, it is Q_M . Differentiating formally at zero gives (6.3). Equivalently, in (3.4), the marker degree counts occurrences of actions above M . The terms of marker degree at least two have absolute mass bounded by $A_{n,M}^{(2)}$, proving (6.4). If the sole tail action lies outside the stated interval, either L is outside a fixed small neighborhood of j_n , or the largest excess is not within a fixed small relative distance of L . For sufficiently large n , the tail action is the unique largest action on the complement of these exceptional events: all other actions are at most the fixed M . Lemma 5.2 gives the final assertion. $\square$

The expression (6.3) is formal. We never integrate the entire infinite tail on a positive circle, where it need not converge. Each individual coefficient uses only $j \leq n$.

6.1 A finite majorant for each core coefficient

The next elementary coefficient estimate avoids a signed Cauchy-integral argument and provides a useful certificate for an individual one-tail term.

Lemma 6.2 (Analytic multiplier estimate). *Let A, H be analytic at zero, real on a real neighborhood, with $A(0) = H(0) = 1$ and $H'(0) = -\beta$. For a fixed analytic logarithm, set*

$$B(u) = \frac{\log H(u) + \beta u}{u^2}, \quad A^*(t) = \sum_{\ell \geq 0} |[u^\ell]A(u)|t^\ell, \quad B^*(t) = \sum_{\ell \geq 0} |[u^\ell]B(u)|t^\ell.$$

Suppose $L_j = aj^2 + (b - \beta)j + d > 0$, $k \geq 0$, and $t = k/L_j$ is inside the convergence radii of both majorants. Then

$$\left| \frac{[u^k]A(u)H(u)^j e^{(aj^2+bj+d)u}}{L_j^k/k!} - 1 \right| \leq A^*(t)e^{jt^2B^*(t)} - 1. \quad (6.5)$$

In particular, uniformly for $n/\log n \leq j \leq 3n/\log n$, $k = n - j$,

$$[u^k]A(u)H(u)^j e^{(aj^2+bj+d)u} = \frac{(aj^2)^k}{k!} e^{(b-\beta)k/(aj)} (1 + O((\log n)^3/n)). \quad (6.6)$$

Proof. Write the expression as $e^{L_j u} G_j(u)$, where $G_j = A \exp(ju^2 B)$ has constant term one. If $g_{j,\ell} = [u^\ell]G_j$, then the normalized coefficient is exactly

$$\sum_{\ell=0}^k g_{j,\ell} \frac{(k)_\ell}{L_j^\ell}.$$

Here $(k)_\ell \leq k^\ell$. The absolute coefficient majorant of G_j at t is at most $A^*(t) \exp(jt^2 B^*(t))$, proving (6.5). In the stated range, $t = O((\log n)^2/n)$ and $jt^2 = O((\log n)^3/n)$. Finally,

$$\left(\frac{L_j}{aj^2} \right)^k = \exp\left(\frac{b-\beta}{a} \frac{k}{j} + O(k/j^2) \right), \quad k/j^2 = O((\log n)^2/n).$$

This proves (6.6). The bound also covers $k = 0$, when $t = 0$ and the coefficient is exactly one. $\square$

6.2 Proof of the main theorem

In Lemma 6.2, use $A = A_M$ and $H = Q_M/u$. Equations (6.2) give its hypotheses. For the finite central range of j , the coefficient in (6.3) is therefore

$$[u^n]V_M^{(1)} = - \sum_{n/\log n \leq j \leq 3n/\log n} T_{n,j} e^{\gamma(n-j)/j} (1 + O((\log n)^3/n)) + O(S_n e^{-cn}). \quad (6.7)$$

The principal summands are positive before the common minus sign, and the relative error is uniform. Lemma 4.1 gives

$$[u^n]V_M^{(1)} \sim -S_n e^{\gamma r_n}. \quad (6.8)$$

Choose $A > |\gamma|/2 + 2$ in Theorem 5.1, and then choose its fixed M . The remainder in (6.4) is $o(S_n e^{\gamma r_n})$, since $r_n \sim \frac{1}{2} \log n$. The analytic coefficient $[u^n]Q_M$ is bounded by C_M^n , while $S_n^{1/n} \rightarrow \infty$; it is negligible on the same scale. Equation (2.6) follows. Its right-hand side is strictly negative, proving eventual negativity. Corollaries 2.2 and 2.3 follow immediately. $\square$

Remark 6.3 (Why the two exponential corrections differ). The forward core inserts $G(q) = q + \beta q^2 + \dots$ inside $e^{aj^2 G(q)}$, generating a term of order $\beta r_n^2/a$. The inverse core inserts $Q_M(u) = u(1 - \beta u + \dots)$ into a large power $Q_M(u)^j$, generating $-\beta r_n/a$. These are different mechanisms. A formal inverse identity, without a tail estimate, does not license replacing one by the other.

7 Universal cancellation and what it does not imply

In the pure model, combine Corollary 2.2 with the forward quadratic equivalent proved in the source article [3]:

$$u_n \sim S_n e^{(1+1/a)r_n^2}. \quad (7.1)$$

This is the only consequence in the present report that imports a nonclassical asymptotic theorem rather than proving its input anew. It gives

$$\boxed{-\frac{v_n}{u_n} \sim \exp\left(-(1+1/a)(r_n^2 + r_n)\right)}. \quad (7.2)$$

The factor tends to zero faster than every fixed power of n^{-1} , because $r_n \sim \frac{1}{2} \log n$. Nevertheless, its logarithm is $o(n)$. Thus the forward and inverse sequences can have identical exponential coefficient type on a factorial–logarithmic scale while differing by a factor invisible to that invariant. This is a concrete obstruction to using type preservation as a substitute for sharp inversion.

There is also a hierarchy of finite-prefix effects. Altering λ_1 or w_2 changes β , and therefore changes the leading inverse coefficient by a factor $\exp(-\Delta\beta r_n/a)$. Altering any other finite set of primitive data, while keeping β unchanged, changes the coefficient only by a relative $o(1)$. The core contains every exceptional primitive datum, but only its quadratic jet survives at this precision.

The tail parameter b and the core parameter β occur through the single combination $b - \beta$. This is not an arbitrary normalization: $bj u$ from the tail and $-\beta j u$ from $Q_M(u)^j$ have the same size at the coefficient saddle. The constant tail term du , and all higher core jets, vanish in the relative leading asymptotic.

None of these results asserts convergence of the original kernel at positive arguments. If $q, u > 0$, the eventually quadratic summands $q^j e^{(aj^2+bj+du)}$ fail even to tend to zero. All uses of the countably infinite equation in the proof are formal and coefficientwise; all analytic evaluations use a fixed finite core.

8 The inverse Borel transform

This section proves Theorem 2.4. It also explains why an entire Borel transform does not by itself provide a Borel sum in a specified direction.

8.1 Envelope identities

For real $x > 0$, continue the saddle coordinate using (2.4), and write

$$E(x) = \frac{xr(x)}{1+r(x)}(1+2r(x)), \quad D(x) = 1+4r(x)+2r(x)^2.$$

Differentiating the defining equation of r gives

$$r'(x) = \frac{r(x)(1+r(x))}{xD(x)}. \quad (8.1)$$

The stationary-action identity in Lemma 4.1, or direct differentiation, then gives

$$E'(x) = 2r(x), \quad E''(x) = \frac{2r(x)(1+r(x))}{xD(x)} > 0. \quad (8.2)$$

These identities preserve every slowly varying term needed below.

By Theorem 2.1 and Stirling's formula,

$$\left(\frac{|v_n|}{n!}\right)^{1/n} \sim \frac{e^{2r_n+1/(1+r_n)}}{n} = \frac{ae^{1/(1+r_n)}}{r_n(1+r_n)} \rightarrow 0.$$

This proves that $\mathcal{B}\hat{Q}$ is entire. In contrast, $|v_n|^{1/n} \sim e^{2r_n-1+1/(1+r_n)} \rightarrow \infty$, so the original formal inverse has radius of convergence zero.

8.2 A second, positive coefficient saddle

For $z > 0$, define

$$\Phi_z(x) = E(x) - x \log x + x + x \log z. \quad (8.3)$$

Equations (8.1)–(8.2) imply

$$\Phi'_z(x) = 2r(x) - \log x + \log z = \log \frac{az}{r(x)(1+r(x))}, \quad (8.4)$$

$$\Phi''_z(x) = -\frac{2r(x)+1}{xD(x)} < 0. \quad (8.5)$$

Thus there is a unique real maximum. If $R(R+1) = az$, it is at

$$m = m(z) = ze^{2R} = \frac{R(R+1)}{a}e^{2R}. \quad (8.6)$$

At this point,

$$\Phi_z(m) = \frac{R}{a}e^{2R}, \quad \sigma_B^2 = \frac{1}{-\Phi''_z(m)} = \frac{m(1+4R+2R^2)}{2R+1}. \quad (8.7)$$

In particular, $\sigma_B \rightarrow \infty$ and $\sigma_B/m \rightarrow 0$.

Apply Stirling's formula to the reference sum

$$H(z) = \sum_{n \geq 1} \frac{S_n e^{\gamma r_n}}{n!} z^n. \quad (8.8)$$

Uniformly for n near m , its summand is

$$\frac{e^{\gamma r_n}}{\sqrt{2\pi n D_n}} e^{\Phi_z(n)} (1 + O(1/n)).$$

The logarithmic prefactor varies by $o(1)$ over a fixed number of standard deviations. On $[m/2, 2m]$, differentiation of (8.5) gives $\Phi'''_z(x) = O(1/(m^2 R))$. Consequently its standardized cubic coefficient is $O(\sqrt{R/m}) \rightarrow 0$. For example, on $|n - m| \leq \sigma_B(m/R)^{1/12}$, the cubic remainder is uniformly $o(1)$. Taylor expansion and the Gaussian Riemann sum therefore give

$$\begin{aligned} H(z) &\sim \frac{e^{\gamma R}}{\sqrt{2\pi m(1+4R+2R^2)}} e^{\Phi_z(m)} \sqrt{2\pi \sigma_B^2} \\ &= \frac{e^{\gamma R}}{\sqrt{2R+1}} \exp\left(\frac{R}{a}e^{2R}\right). \end{aligned} \quad (8.9)$$

For completeness, the omitted tails cannot change this equivalent. Within $[m/2, 2m]$, the second derivative is bounded above and below by negative constant multiples of $1/(mR)$, giving a Gaussian upper bound outside the central window. At either endpoint the phase has already lost at least cm/R . Outside that interval, concavity bounds the phase by the tangent at the endpoint. Its outward slope has absolute value at least c/R , so the right tail is bounded

by a geometric sum. Polynomial factors from $e^{\gamma r_n}/\sqrt{nD_n}$ are absorbed by the same bound; on the finite left tail their logarithm is $O(\log m)$, negligible beside m/R . These observations justify the sum asymptotic without an unproved interchange of a nonuniform expansion and an infinite sum.

Finally, for every $\epsilon > 0$, Theorem 2.1 gives a fixed N_ϵ such that

$$(1 - \epsilon)S_n e^{\gamma r_n} \leq -v_n \leq (1 + \epsilon)S_n e^{\gamma r_n} \quad (n \geq N_\epsilon).$$

The finite initial part of the Borel series is a polynomial in z , negligible compared with (8.9). Positivity of the reference summands now sandwiches $-\mathcal{B}\hat{Q}(z)$ between $(1 \pm \epsilon)H(z)$, up to negligible terms. Letting $\epsilon \rightarrow 0$ proves (2.9).

8.3 The positive-direction obstruction

For a positive Laplace parameter t , the ordinary factorial-normalized Borel–Laplace prescription would be

$$\frac{1}{t} \int_0^\infty e^{-z/t} \mathcal{B}\hat{Q}(z) dz.$$

The equivalent (2.9) shows that its integrand is eventually negative and has logarithm of absolute value

$$\frac{R(z)}{a} e^{2R(z)} - \frac{z}{t} + O(R(z) + \log R(z)).$$

Since $R(z) \sim \sqrt{az}$, this tends to positive infinity. The improper integral diverges to $-\infty$. There is no oscillatory cancellation on this ray. This proves the final assertion of Theorem 2.4; it says nothing against summation in another direction, nor against a different acceleration procedure.

9 The smallest formal inverse term

Fix $0 < t < 1$ and put $R = \frac{1}{2} \log(1/t)$. The reference term associated with Theorem 2.1 is

$$h_t(x) = \frac{e^{\gamma r(x)}}{\sqrt{D(x)}} \exp(E(x) + x \log t). \quad (9.1)$$

The large phase $\Psi_t(x) = E(x) + x \log t$ is strictly convex by (8.2); its derivative is $2(r(x) - R)$. Its unique minimum is therefore attained at

$$m = m(t) = \frac{R(R+1)}{at}, \quad \Psi_t(m) = -\frac{R^2}{at}. \quad (9.2)$$

Moreover

$$\Psi_t''(m) = \frac{2R(R+1)}{m(1+4R+2R^2)} \sim \frac{1}{m}. \quad (9.3)$$

The derivative of the logarithmic prefactor in (9.1) is $O(1/m)$ near m . Its inclusion shifts the continuous reference minimum by only $O(1)$; rounding to a neighboring integer changes the minimum value by $1 + o(1)$. To see that this is the global minimum, note that, for all sufficiently large x , the second derivative of $\log h_t(x)$ is $E''(x) + O(x^{-2}) > 0$. On any fixed bounded range of positive indices the reference value is only a fixed power of t , whereas $h_t(m)$ is exponentially smaller. It follows that

$$\min_{n \geq N} h_t(n) \sim \frac{\exp(-R^2/(at) + \gamma R)}{\sqrt{1+4R+2R^2}}. \quad (9.4)$$

We must still justify replacing the reference coefficients by the exact ones inside a minimum. For every fixed $t > 0$, $|v_n|t^n \rightarrow \infty$, because $|v_n|^{1/n} \rightarrow \infty$; hence a minimum over $n \geq N$ exists. As $t \downarrow 0$, any minimizing index tends to infinity: every fixed finite set of terms is polynomially small in t , but the term near $m(t)$ is smaller than every fixed power of t . On all sufficiently large indices, the exact-to-reference ratio lies uniformly in $[1 - \epsilon, 1 + \epsilon]$. The same sandwich argument used for the Borel sum now applies to the minimum. It proves (2.10).

For the index assertion, on the scale $n = m + y\sqrt{m}$, Taylor's formula and (9.3) give, uniformly for bounded real y ,

$$\log \frac{h_t(n)}{h_t(m)} = \frac{y^2}{2} + o(1). \quad (9.5)$$

For any fixed $\delta > 0$, eventual convexity gives a strictly positive logarithmic gap outside $|n - m| < \delta\sqrt{m}$, first locally by (9.5) and then globally by monotonicity on each side of the reference minimizer. The uniform relative error in the exact coefficients tends to zero, so it cannot close this fixed gap. Thus $|n_{\min} - m| < \delta\sqrt{m}$ eventually. Since δ is arbitrary, (2.11) follows.

Remark 9.1 (Three distinctions about optimal truncation). The smallest coefficient term is not automatically the error of a truncated analytic function. Such a conclusion requires a specified analytic realization and a remainder theorem. Also, an equivalent with unspecified relative $o(1)$ error does not determine the minimizing integer to $O(1)$, because the curvature is only of order $1/m$. Finally, finite initial zero coefficients would make an unrestricted minimum equal to zero; this is why the theorem is stated after an eventual-negative index N . None of these issues is hidden by the notation “least term.”

10 Exact computations and reproducibility

10.1 An integer-normalized recurrence

The program `code/verify.py` computes $V_n = n!v_n$ and $P_{j,n} = n![u^n]\widehat{Q}(u)^j$. Differentiating the power gives the exact recurrence

$$P_{j,n} = j \sum_{k=1}^{n-j+1} \binom{n-1}{k-1} V_k P_{j-1,n-k} \quad (j \geq 2), \quad (10.1)$$

with $P_{1,n} = V_n$, $V_1 = 1$, and vanishing coefficients below the natural degree. The inverse kernel then gives, for $n \geq 2$,

$$\begin{aligned} V_n = & - \sum_{k=1}^{n-1} \binom{n}{k} V_k \lambda_1^{n-k} \\ & - \sum_{j=2}^n w_j \sum_{k=j}^n \binom{n}{k} P_{j,k} \lambda_j^{n-k}. \end{aligned} \quad (10.2)$$

The right-hand side only contains previously available data. Integer primitive parameters give integers throughout, so there is no roundoff in the coefficient files. The algorithm uses polynomially many arithmetic operations, although the cost of large-integer arithmetic grows with the size of these integers. No unit-cost bit-complexity claim is intended.

10.2 Independent finite checks

The supplied run used five models: $\lambda_j = j^2$; $\lambda_j = 2j^2$; $\lambda_1 = 0$ with $\lambda_j = j^2$ for $j \geq 2$; $\lambda_j = j^2 + j$; and $w_2 = 2$ with all other weights one and $\lambda_j = j^2$. The main pure model was computed through degree 320, and each additional model through degree 160.

There were 255 exact assertions in the recorded run. For each model, the integer recurrence was compared through degree 14 with the independent finite partition formula (3.4). The marked one-tail formula was separately checked through degree 12 at cutoffs $M = 1, 2, 3$ against exact truncated-polynomial evaluation of the analytic-core response. The small cutoff checks test the exact identity, not the asymptotic sufficiency of those particular cutoffs. All assertions passed. The check program does not serve as a machine proof of the asymptotic localization theorem.

The files `data/*_exact_egf.csv` contain $V_n = n!v_n$, not v_n . This normalization is recorded explicitly to prevent a factorial from being mistaken for a disagreement in coefficient growth. The file `data/verification.json` records the check count and run parameters.

10.3 Asymptotic diagnostics

Table 1 compares the exact pure-model coefficients with the proved equivalent. Decimal evaluation uses 90-digit `mpmath` arithmetic. These numbers are diagnostic approximations, not intervals.

Table 1: Pure model $a = 1$: exact coefficients divided by the proposed and now proved leading equivalent. The slow approach is visible.

n	r_n	$v_n/(-S_n e^{-2r_n})$
20	1.087760	0.448613578
50	1.368199	0.649597330
100	1.593244	0.757155401
160	1.751348	0.813338189
200	1.827836	0.835909537
240	1.890976	0.852577838
300	1.969011	0.870968827
320	1.991732	0.875899988

The agreement of the rows at $n = 50, 100, 200, 300$ with the earlier source’s conjectural diagnostic table [3] is a useful normalization check. The ratio at degree 320 is still about 0.876, so the leading equivalent is not a high-accuracy finite-order approximation at these degrees. The theorem claims a limit, not a certified onset value.

For the four other models, the ratios at degree 160 to $-S_n e^{(b-\beta)r_n/a}$ are respectively about 0.84676, 0.76341, 0.78906, and 0.84791, in the order listed after the main model above. These are useful tests of the core jet, the slope coefficient, and the constant in the exponent; a finite table cannot prove the universal perturbation theorem.

10.4 Rebuilding the package

From the package directory, run

```
python code/verify.py -degree 320
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third LaTeX pass may be needed after changing the table of contents. The exact checks use Python’s standard-library integers and rational arithmetic. Only the diagnostic decimal evaluations require `mpmath`. The program performs no network requests and does not require a local ProveIt checkout.

11 Proof audit and a formalization route

11.1 Dependencies and scope

The main coefficient theorem depends on the exact formal configuration identity, elementary analytic implicit inversion for a finite core, an absolute background-cloud bound, and a one-dimensional discrete saddle estimate. No forward coefficient asymptotic or conjectural source theorem is used in its proof. The two Borel and least-term theorems depend only on the proved inverse equivalent and elementary saddle analysis.

The forward-to-inverse ratio (7.2) additionally uses (7.1) from the repository. Its provenance is separated because proving the inverse conjecture does not retroactively formalize or independently audit every forward result in its source.

The proof's most delicate step is Theorem 5.1. Its successive bounds are stated on progressively smaller sets: macroscopic excess and a unique large action, a fine excess window, a submesoscopic background, and finally a polynomial cloud. At each removal, the mass is estimated before cancellation. The last estimate retains a cutoff-dependent factor e^{-2Mr_n} ; the positive cloud constant does not depend on M . This independence is what permits an arbitrarily accurate fixed cutoff. An $e^{o(n)}$ estimate alone would not suffice.

No all-order asymptotic expansion, numerical threshold for the asymptotic regime, smallest sufficient inverse core, or open-sector Borel continuation is claimed. The proofs are conventional mathematical proofs supplied for independent checking; the report contains no Lean code and no claim of a completed formal verification.

11.2 A modular route to Lean

A practical formalization should not start by encoding an infinite saddle sum all at once. The finite algebraic portion can be separated completely from the asymptotics.

First, formalize the coefficientwise recursion for $\hat{U}$, its compositional inverse, and the marker derivative of a finite implicit kernel. The recurrence (10.2) and finite partition formula give two executable representations that can be compared at arbitrary finite truncation orders. Formalizing their equality is stronger than checking selected coefficients.

Second, state the majorant theorem for finite weighted configurations of a fixed total excess. Its essential ingredients are the fusion inequality, the factorial-ratio bound, and a marked exponential generating-function inequality. This finite theorem can expose all constants and avoid Hahn series entirely. The countably infinite model is recovered coefficient by coefficient.

Third, formalize a reusable discrete Laplace lemma for a strictly concave phase with a diverging Gaussian width, uniform local Taylor bounds, and explicit tail inequalities. The action saddle and Borel saddle then become two instances with different curvature scales. A corresponding convex minimum lemma handles the least term.

Finally, connect the coefficient field to the repository's transseries support API by substituting $q = e^{-x}$. This last bridge should state only the formal series object and its coefficient theorems. An analytic realization requires its own theorem and should not be silently attached to the formal representation.

12 Further research questions

The questions below are continuations not resolved by this report. The first three are the most direct extensions of the proof; the later ones connect sharp coefficients to summation, effective computation, and transseries algebra.

Research question 1 (The smallest sufficient inverse core). For the pure quadratic model, is a two-action core already sufficient for a full multiplicative inverse equivalent? More generally,

characterize the smallest M for which the first marked response has the same leading coefficient as the inverse. The proof here chooses a possibly large cutoff to overcome a deliberately coarse absolute majorant. That choice is an existence result, not a sharp threshold. A smaller cutoff likely requires retaining cancellations between additional tails instead of estimating those tails absolutely.

Research question 2 (A complete inverse large-order expansion). Determine an all-order expansion of $v_n/(-S_n e^{\gamma r_n})$ in explicit functions of r_n and n^{-1} . The next terms receive contributions from the cubic jet of the inverse core, the constant tail term d , the Stirling correction, and the displacement of the action saddle caused by $e^{\gamma(n-j)/j}$. Their relative sizes must be ordered before truncation. A useful initial target is a first correction that explains the visible error in Table 1, with a proved uniform remainder and no empirically fitted constants.

Research question 3 (Subquadratic signed condensation). For $\lambda_j = aj^p$, $1 < p < 2$, develop the analogue of the absolute localization theorem at a scale fine enough to survive cancellation. At the bare saddle, the inverse-core perturbation $j \log(Q_M(u)/u)$ can contain several terms that do not tend to zero. Identify the necessary core jet order and determine the resulting cancellation exponent. The forward finite-core threshold cannot simply be copied: forward and inverse cores enter different large powers.

Research question 4 (Slowly varying departures from a quadratic tail). Replace the eventual polynomial identity by $\lambda_j = aj^2 + bj + \eta j / \log j + o(j / \log j)$, with suitable control of local oscillations. Since $(n-j)/(j \log j) \rightarrow 1/2$ at the bare saddle, the direct one-tail calculation predicts an extra factor $e^{\eta/(2a)}$. Prove a uniform fusion bound for this class and determine which finite-difference assumptions are actually necessary. More irregular perturbations could destroy a smooth local saddle while preserving a logarithmic coefficient law.

Research question 5 (Effective, rather than asymptotic, cancellation bounds). Make the constants in (5.9), (5.10), and (5.19) explicit for the pure model. Combine them with interval bounds for the saddle sum and the analytic core to produce a computable enclosure for $v_n/(-S_n e^{-2r_n})$. The target is an inequality valid for a stated finite range of degrees, not a floating-point agreement or a nonconstructive “sufficiently large” statement.

Research question 6 (Angular inverse Borel growth). Theorem 2.4 concerns one real ray. Determine the growth of $\mathcal{B}\hat{Q}(z)$ for complex z in sectors, including the transition between constructive and destructive phase interference. Eventual signs reduce the positive ray to a real saddle but provide no lower bound off that ray. A solution could locate directions of ordinary summability or identify a suitable acceleration scale without confusing entire continuation with acceptable growth.

Research question 7 (Least term versus an actual analytic remainder). For a specified negative-direction analytic realization of the inverse, determine whether the scale in (2.10) is attained by its optimal truncation error, or whether the error has an additional factor. This requires both an upper remainder estimate and a matching lower bound. The coefficient theorem alone gives neither. Any comparison should specify the sector, the branch, and whether the truncation index is chosen continuously or integrally.

Research question 8 (A moving-saddle asymptotic algebra). Construct an algebra of coefficient asymptotics that retains an implicit slowly varying coordinate such as $r(1+r)e^{2r} = an$, rather than only a fixed factorial scale. Determine composition, inversion, differentiation, and multiplication rules that preserve enough information to recover (7.2). A useful goal is an effective analogue of the factorial-asymptotic calculus in [6], but with a nontrivial invariant for the zero-factorial-type sequences considered here.

Research question 9 (Robustness under signed or oscillatory primitive data). How far can the assumptions $w_j, \lambda_j \geq 0$ be relaxed? Finitely many signed primitive weights may still admit a controlled absolute majorant, but the quadratic core jet can vanish or change sign, and oscillatory tail weights may introduce several equally dominant saddles. Classify the cases with eventual inverse sign, periodic sign patterns, or persistent cancellation at leading order. This question requires a new statement rather than a formal replacement of positive constants by absolute values.

13 Conclusion

The sharp quadratic inverse conjecture is reduced to, and proved by, an absolute localization estimate strong enough to survive the signed cancellation. A fixed finite core incorporates every exceptional primitive parameter, yet only its quadratic coefficient survives in the leading inverse law. The same law determines an entire Borel transform whose positive-axis growth is too large for ordinary Laplace summation, and a precise smallest-formal-term scale.

The principal mathematical distinction is between preserving a coefficient class and preserving a coefficient equivalent. In this model the difference is a factor of size $\exp(-c(\log n)^2)$ relative to the forward series. Capturing that factor requires summing the analytic-core cancellations and bounding the omitted signed configurations on the smaller scale, not merely transferring a Gevrey estimate.

References

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